Machine Learning and Pattern Classification

Task 3: Data Exploration

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Abstract

I address all required parts of Task 3: Data Exploration for the domestic sound event detection dataset. I reviewed assigned recordings in Label Studio, quantified annotator agreement from the time-aligned annotation arrays, derived binary labels for each time step, and analyzed label characteristics, metadata, feature statistics, correlations, and a PCA-based feature-space visualization. The dataset is suitable for the next classification phase, but class imbalance, room bias, metadata noise, and temporal ambiguity remain important challenges.

1 Introduction

Task 3 focuses on exploratory data analysis before model training. I therefore structure the report exactly like the assignment: (1) Verify Annotations & Qualitative Analysis, (2) Annotations, (3) Audio Features, and (4) Conclusions. The development package contains `metadata.csv`, `annotations.csv`, and one `.npz` file per recording. Each feature file stores time-aligned acoustic features, metadata, and an annotation tensor of shape $[T, C, A]$, where T is the number of segments, C the number of classes, and A the number of annotators.

My final notebook processed all 3,656 development recordings and 37,419 annotation rows. The dataset contains 369 device labels, 408 collectors, and 602 annotators.

2 Verify Annotations & Qualitative Analysis

2.1 Review Procedure

I reviewed the assigned recordings in Label Studio using the *Review* interface and *Compare All*. For each file, I checked the metadata, listened to the recording, and compared the annotations in summary and side-by-side view. I then checked whether the selected classes were plausible, whether temporal boundaries were reasonable, and whether repeated actions were split consistently. If an annotation contained a clear mistake, I corrected it with *Fix + Accept*; otherwise, I used *Accept*. Small onset and offset differences were not treated as errors when both versions remained plausible.

Reviewed files and applied fixes. The reviewed files show the main correction types in practice: class relabeling, adding missed events, removing unsupported regions, and refining temporal boundaries.

- `002635.wav`: class correction only. I changed `bell_ringing` to `phone_ringing`; no boundary changes were needed. Agreement remained 71.58% because the other two annotations were already consistent.
- `003862.wav`: strong agreement case. All annotations marked `toilet_flushing` and `running_water`; only small timing differences remained, so I accepted all versions without changes. Agreement was 88.71%.
- `003863.wav`: missed short events. I added a missing `keychain` event in one annotation and a missing `footsteps` event in another. Agreement for this review was 41.01% before correction, showing that short events were easy to miss.
- `003864.wav`: plausible segmentation variants. All annotations used the same class set, but repeated kitchen sounds were grouped differently. I accepted the annotations without changes because multiple segmentations remained plausible. Agreement was 74.15%.
- `003865.wav`: minor boundary differences only. The class inventory and event count were already consistent, so I accepted all annotations without edits. Agreement was 78.95%.
- `000143.wav`: class confusion and typing segmentation. I changed `bell_ringing` to `phone_ringing` for 90903455, split `keyboard_typing` for 90776591, merged adjacent typing for 90594374, and left 90539059 unchanged. Agreement increased from 55.52% to 82.59%.
- `000144.wav`: missing and inaccurate regions. I added a missing `footsteps` event for 90896328, removed one unsupported `footsteps` region and moved `bell_ringing` earlier for 90778696, and shortened `footsteps` for 90777807. Agreement increased from 65.59% to 77.34%.
- `000146.wav`: boundary correction only. I shortened the overextended `door_open_close` region for 90778999, 90767272, and 90585649. Agreement increased from 86.38% to 94.69%.
- `000148.wav`: minor timing refinement. I adjusted the boundaries of `window_open_close` and `footsteps` for 90780235. Agreement increased from 88.16% to 94.79%.
- `000149.wav`: overlap and boundary correction. I slightly extended `footsteps` for 90780606 and 90746387, and removed one short duplicate `wardrobe_drawer_open_close` region for 90572339. Agreement increased from 78.30% to 85.36%.

2.2 Summary of Disagreements

Across the reviewed files, I observed four main disagreement patterns. First, annotators sometimes placed slightly different onset and offset boundaries for the same event, especially for short transients and open/close actions. Second, repeated events were segmented differently: some annotators merged footsteps or repeated kitchen sounds into broad regions, while others split them into several shorter regions. Third, I observed class confusion between acoustically similar classes, especially `door_open_close` vs. `window_open_close` and `bell_ringing` vs. `phone_ringing`. Fourth, short target events such as `keychain` or brief `footsteps` were sometimes missed completely.

The individual reviews confirm these patterns. In `002635.wav` and `000143.wav`, I corrected class confusion between notification sounds. In `003863.wav` and `000144.wav`, the main problem was missed or overextended short events. In `000146.wav`, `000148.wav`, and `000149.wav`, the classes were already correct and the remaining work was boundary refinement or removal of overlapping duplicate regions. Files `003862.wav`, `003864.wav`, and `003865.wav` showed that some differences were still acceptable and did not require changes.

Not every disagreement was an error. For example, in `003862.wav`, the annotators marked the same two events, `toilet_flushing` and `running_water`, with only small temporal differences, so no correction was needed. In `003864.wav`, all annotators agreed on the class set but differed in how repeated short kitchen events were grouped; these alternative segmentations remained plausible and were accepted.

2.3 Case Studies

Case 1: `002634.wav`. This file had a very low agreement score of 24.49%. The annotators disagreed both on class identity and on segmentation. One annotation used `door_open_close`, while others used `window_open_close`. The walking activity was also grouped differently. After listening and inspecting the spectrogram, I concluded that the object-related sound was `window_open_close`. I removed unsupported extra regions, corrected the label where needed, and refined the `footsteps` segmentation. This case shows that disagreement can go beyond small boundary shifts and include class confusion.

Figure 1: Case study for file `002634.wav`. The figure shows a class-identity disagreement in which the same object-related sound was labeled as `door_open_close` in one annotation and as `window_open_close` after review.

Case 2: `000143.wav`. This file initially combined two disagreement types: class confusion and segmentation disagreement. One annotation used `bell_ringing` instead of `phone_ringing`. In addition, the `keyboard_typing` activity was segmented differently: one annotation split it into two parts, while another merged adjacent typing activity. During review, I corrected the class label and adjusted the segmentation. Agreement increased from 55.52% to 82.59%. This file is a good example of how acoustically similar notification sounds and interrupted typing activity can lead to different annotations.

Figure 2: Case study for file `000143.wav` showing correction from `bell_ringing` to `phone_ringing` and different `keyboard_typing` segmentation.

3 Annotations

3.1 Annotator Agreement

I quantified agreement on the segment level using the time-aligned tensor `annotations[T,C,A]`. For each annotator, class, and segment, I binarized overlap with the rule "overlap > 0" and computed pairwise Jaccard agreement, averaged across files.

The overall mean pairwise segment-level Jaccard agreement was **0.7004**. As a coarse file-level check, the mean pairwise Jaccard similarity between annotator label sets was **0.9105**. Thus, annotators usually agreed on which classes were present in a file, while temporal boundaries were harder to place consistently.

Agreement differed across classes. The highest values were observed for `vacuum_cleaner` (0.8744), `keyboard_typing` (0.8513), and `running_water` (0.8372). The lowest values occurred for `wardrobe_drawer_open_close` (0.5568), `footsteps` (0.5937), and `door_open_close` (0.5948). Sustained and distinctive sounds are easier to annotate consistently than short or ambiguous open/close actions.

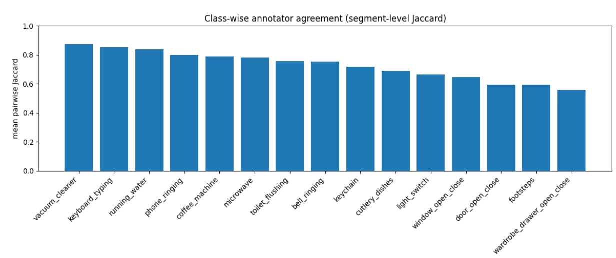

Figure 3: Class-wise segment-level Jaccard agreement across annotators.

3.2 Temporal Labels vs. Uploader Tags

I also compared temporally derived labels to the uploader-provided target tags. The exact file-level set match was **0.8236**, so metadata is helpful but not fully reliable. The strongest agreement was found for `toilet_flushing` (0.9973), `vacuum_cleaner` (0.9959), and `coffee_machine` (0.9970). The weakest match was observed for `footsteps` (0.9248), followed by `wardrobe_drawer_open_close` (0.9768) and `cutlery_dishes` (0.9776).

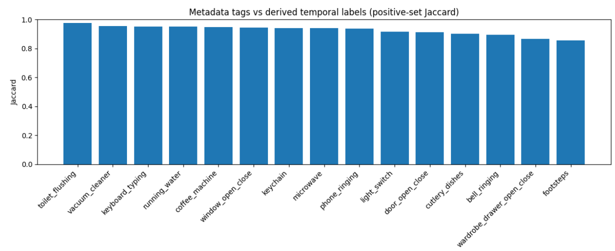

Figure 4: Agreement between uploader target tags and temporally derived labels.

3.3 Converting Annotations to Labels

I derived binary labels for every time step by aggregating across annotators. Let $y_{tca} \in [0, 1]$ be the proportional overlap for segment t, class c, and annotator a. First, I binarized each annotator-specific value:

$$b_{tca} = \mathbb{I}(y_{tca} > 0).$$

Second, I applied majority vote across annotators:

$$\hat{y}_{tc} = \mathbb{I}\left(\frac{1}{A_t}\sum_a b_{tca} > 0.5\right).$$

This method is simple and reproducible, but it assumes equal annotator reliability and may suppress rare valid events if only one annotator detected them.

3.4 Label Characteristics

The most frequent positive segment labels were `footsteps` (31,392 segments, 1,831 files), `running_water` (24,593, 995 files), and `keyboard_typing` (18,626, 728 files). The least frequent were `bell_ringing` (3,182, 278 files), `light_switch` (3,594, 753 files), and `window_open_close` (4,316, 481 files), so the segment labels are clearly imbalanced.

Duration statistics also matched event semantics. The longest mean durations occurred for `vacuum_cleaner` (18.27 s), `microwave` (13.93 s), `coffee_machine` (13.02 s), and `running_water` (11.99 s), while the shortest were `light_switch` (2.32 s), `wardrobe_drawer_open_cl` (3.65 s), and `door_open_close` (3.67 s). The most common co-occurrences were `door_open_close + footsteps` (914 files), `footsteps + light_switch` (533), `footsteps + keychain` (489), and `cutlery_dishes + running_water` (383).

4 Audio Features

4.1 Metadata

Metadata analysis showed a reasonable balance between **1,970 static** and **1,686 mobile** recordings. The most common environments were `kitchen` (1,070), `bedroom` (711), `hallway` (682), `living_room` (557), `office` (430), `bathroom` (364), and `toilet` (199).

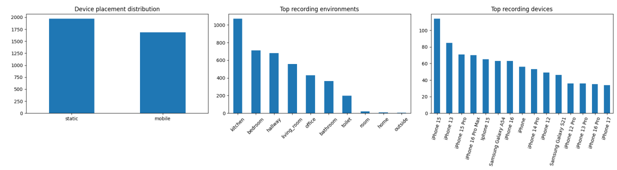

Figure 5: Metadata distributions for device placement, recording environments, and recording devices.

4.2 Feature Statistics

I analyzed mean, standard deviation, and range for the acoustic feature families. The features differ strongly in scale. `power` showed the largest variability (mean 179.04, std 530.79, range 24,860.71), while frequency-based descriptors such as `rolloff_high`, `centroid`, and `bandwidth` naturally had larger absolute values than bounded descriptors such as `flatness` and `zcr`. This mainly reflects different physical meanings and units, but it also motivates standardization before multivariate analysis.

4.3 Feature-Family Correlation

The strongest positive correlations were `mfcc_d` with `mfcc_d2` (0.9628), `zcr` with `centroid` (0.9598), `bandwidth` with `rolloff_high` (0.9516), and `flux` with `energy` (0.9187). I also observed strong negative correlations, especially `melspect` with `mfcc` (-0.8677). Overall, correlation was stronger within related feature groups than across unrelated descriptors.

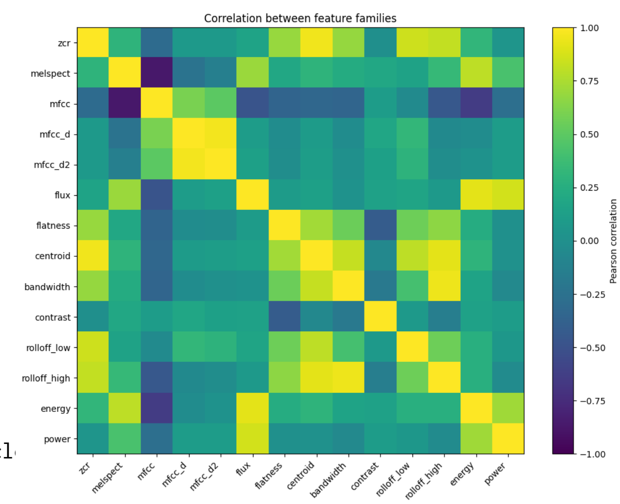

Figure 6: Feature-family correlation visualization.

4.4 Bonus: Feature Space Visualization

For the optional down-projection task, I selected a high-quality subset of **3,750** segments, using **250 single-class, high-agreement segments per class**. After standardization, PCA showed that the first two principal components explained **47.69%** of the variance. The plots suggest partial structure rather than perfect separation: some classes form denser regions, while others overlap because of shared context or similar spectral structure.

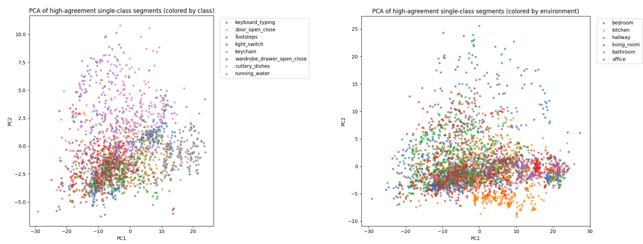

Figure 7: PCA down-projection of the high-quality segment subset. Left: class-colored view. Right: environment-colored view.

5 Conclusions

The dataset is suitable for the next classification phase. The annotations are reasonably consistent, the time-step label derivation is straightforward, and the feature representation captures diverse acoustic properties. However, the next phase will still be challenging because of class imbalance, room bias, metadata noise, device variation, and temporal ambiguity in several classes.

The most important biases are recording-device bias, room imbalance, class-frequency imbalance, and annotation subjectivity. These factors can make some classes easier to learn than others and may encourage a model to rely on context instead of the true event. A domestic sound event dataset can support useful smart-home and accessibility applications, but it also creates privacy risks if used for invasive monitoring. For that reason, dataset limitations and intended use should be documented clearly.

6 Disclosure of LLM and AI Tool Use

I used AI tools to support notebook debugging, report structuring, and language refinement. In particular, I used ChatGPT to help organize the analysis pipeline, clarify method descriptions, and improve the readability of the final report. All generated suggestions, code, and text were manually checked against the task description, the dataset structure, and the final notebook outputs before inclusion in the submission.